

Project Initialization and Planning Phase

Date	July 14, 2026
Team ID	
Project Title	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

Our machine learning-driven system automates credit card approval decisions. By evaluating an applicant's demographic and financial data, the model accurately predicts whether a request should be granted or denied. This approach minimizes human error, accelerates the evaluation timeline, and ensures uniform, reliable outcomes.

Project Overview	
Objective	Construct a highly accurate, AI-powered predictive model to determine credit card application approvals and rejections based on applicant profiles.
Scope	This initiative covers the end-to-end ML lifecycle: gathering and cleaning data, engineering features, and training/evaluating models prior to deployment. We will benchmark several algorithms to select the most efficient one for live web-based predictions.
Problem Statement	
Description	Financial institutions process vast volumes of credit requests daily. Relying on manual reviews is slow, subjective, and susceptible to mistakes, which drives up operational expenses and frustrates applicants with long wait times.
Impact	Transitioning to an automated system drastically speeds up resolutions, heightens accuracy, mitigates default risks, slashes manual workload, and ultimately boosts overall user satisfaction.
Proposed Solution	
Approach	Historical applicant data is pre-processed and used to train Machine Learning models including Logistic Regression, Decision Tree, Random Forest, and XGBoost. The best-performing model is integrated into a Flask web application for real-time prediction.
Key Features	• Automated Credit Card Approval Prediction • High Accuracy using Machine Learning • Real-Time Prediction through Web Application • User-Friendly Interface • Fast Decision-Making Process • IBM Watson Machine Learning Deployment Support • Secure and Scalable Architecture

Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor (4+ Cores)
Memory	RAM specifications	8 GB RAM (Minimum), 16 GB Recommended
Storage	Disk space for datasets, trained models, and application files	512 GB SSD / 1 TB HDD
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost
Development Environment	IDE	Jupyter Notebook, Visual Studio Code, PyCharm
Data		
Data	Source, Size, Format	Credit Card Approval Dataset (Kaggle), CSV Format, ~20,000+ Applicant Records